

Hybrid AC/DC Module Technical Notebook

Current C++ Architecture, Information Models, Mathematical Models, and Solver Workflows

HACDCPF C++ Module

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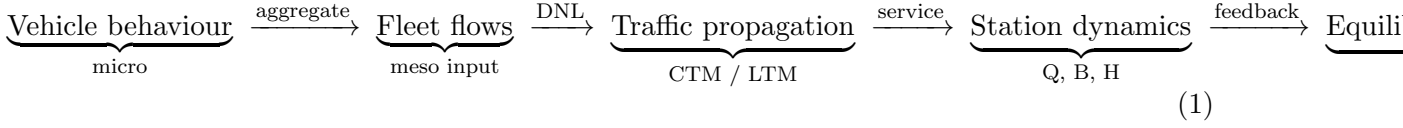
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1 Traffic System Analysis: A Meso-Micro Simulation Framework for Mixed Fuel and Electric Vehicles

Remark 1 (Implementation target). *This section is the mathematical contract for the `hacdcp f::evpt` traffic sub-module. Function signatures, data structures, and algorithmic steps given here are the specification that the C++ implementation in `src/ev_power_traffic/` follows. Where a function is already implemented a cross-reference to the source file is given; where it is planned, the label **Planned** appears.*

1.1 Model Architecture

The framework is organised as a five-layer pipeline:



Each layer has a precisely defined input–output contract:

- *Vehicle behaviour layer*: answers how individual vehicles choose routes and charging plans given perceived travel times, queue lengths, and energy prices.
- *Fleet aggregation layer*: converts large populations of identically-behaving vehicles into continuous OD flows, making Dynamic Network Loading (DNL) tractable.
- *CTM/LTM layer*: propagates fleet flows through the road network, computing shockwaves, spillback, travel times, and FIFO-consistent trajectories.
- *Station service layer*: models charging-station queues, plug utilisation, service completions, and dwell populations as endogenous nodes inside the network loading.
- *Equilibrium/optimisation layer*: closes the loop; at equilibrium, no vehicle has an incentive to deviate (Wardrop I for ICVs, economic UE for EVs).

This section covers:

- micro ICV and EV behaviour models (route choice, fuel consumption, SOC dynamics, charging and V2G decisions);
- fleet aggregation from micro behaviour to OD flow variables;
- CTM-based system-level network loading with cell-wise spatial state, route-commodity tracking, and DUE via MSA or variational inequality methods; system-optimal dynamic traffic assignment is formulated as an LP under triangular-FD assumptions;
- LTM-based meso-micro loading with cumulative curves, FIFO trajectory reconstruction, and an individual EV interaction layer;
- charging stations as endogenous service nodes;
- the dynamic user-equilibrium fixed-point problem and MSA solution;
- system-optimal LP, MILP infrastructure design, and MPC extensions.

This section does not cover:

- optimal power flow or grid dispatch (Section ??);
- carbon-flow or resilience analysis;
- full agent-level simulation of every vehicle.

1.2 Basic Sets, Indices, and Network Definition

1.2.1 Time horizon

The discrete time set is $\mathcal{K} = \{0, 1, \dots, T\}$ with uniform step size Δt (hours). A flow expressed in veh/h is multiplied by Δt to obtain vehicles per step.

1.2.2 Road network

The road network is a directed graph $\mathcal{G}^R = (\mathcal{N}^R, \mathcal{A}^R)$, where $\mathcal{N}^R$ is the set of road nodes and $\mathcal{A}^R$ the set of directed road links. Each link $a \in \mathcal{A}^R$ is characterised by:

$$\begin{aligned} L_a & \text{ length (km),} \\ v_a^{ff} & \text{ free-flow speed (km/h),} \\ w_a & \text{ backward wave speed (km/h),} \\ \bar{q}_a & \text{ capacity (veh/h),} \\ N_a^{jam} & \text{ jam storage (veh).} \end{aligned}$$

The triangular fundamental diagram (FD) is

$$q_a(k_a) = \begin{cases} v_a^{ff} k_a & k_a \leq k_a^c, \\ w_a(k_a^j - k_a) & k_a > k_a^c, \end{cases} \quad (2)$$

with critical density $k_a^c = \bar{q}_a / v_a^{ff}$ and jam density $k_a^j = \bar{q}_a(1/v_a^{ff} + 1/w_a)$. The backward wave speed satisfies

$$w_a = \frac{v_a^{ff} k_a^c}{k_a^j - k_a^c}. \quad (3)$$

1.2.3 OD pairs, vehicle classes, and demand

The OD set is $\mathcal{W}$; a pair $w = (o, d) \in \mathcal{W}$. The vehicle class set is $\mathcal{C} = \{F, E\}$ (fuel vehicles and EVs).

Full DUE mode (departure time endogenous). Each OD pair w has a fixed total demand D_w over a departure window $\mathcal{K}_w^{dep} \subseteq \mathcal{K}$ and a desired arrival time t_w^* . Departure time is an endogenous choice variable. Given EV penetration $p_w^{EV} \in [0, 1]$:

$$D_w^F = (1 - p_w^{EV}) D_w, \quad D_w^E = p_w^{EV} D_w. \quad (4)$$

The departure profile $d_{w,k} = \sum_r h_{w,r,k}^F + \sum_\omega h_{w,\omega,k}^E$ is an equilibrium *output*, not a given. Unless stated otherwise, *DUE refers to this full mode*.

Fixed-departure mode (departure time exogenous). When an exogenous departure profile $d_{w,k}$ is supplied, the model reduces to a fixed-departure dynamic assignment: equilibrium is imposed only across routes or activity plans *within* each departure step k . In this case the demand split is

$$d_{w,k}^F = (1 - p_{w,k}^{EV}) d_{w,k}, \quad d_{w,k}^E = p_{w,k}^{EV} d_{w,k}. \quad (5)$$

This is a restricted special case of the full DUE; benchmark tests in §1.15 use fixed-departure mode for comparability with published results.

1.2.4 Route and activity-plan sets

The ICV route set for OD w is $\mathcal{R}_w$. An EV *activity plan* $\omega = (r, s, \chi)$ combines a road path r , a *single* optional charging stop $s \in \mathcal{S} \cup \{\emptyset\}$, and a charging/V2G strategy χ (charging power profile, charging energy, discharging energy, dwell duration). The current implementation supports single-stop plans. Multi-stop plans (secondary charging after V2G discharge) are a planned rolling-horizon extension; until then, secondary charging after V2G must be handled by including a separate plan with the net energy already accounting for the discharge. The full activity-plan set for OD w at step k is $\Omega_{w,k}$. ICVs choose a route; EVs choose an activity plan.

In implementation, $\Omega_{w,k}$ is generated by: (i) computing a K -shortest-path set $\mathcal{R}_w^K$ for each OD pair; (ii) enumerating stations reachable along each candidate path; (iii) generating charging templates χ (no-stop, charge-only, discharge-only, sequential charge-then-discharge); (iv) pruning all plans that violate the SOC feasibility constraint (16); and (v) retaining the feasible subset for assignment. Formally,

$$\Omega_{w,k} = \{(r, s, \chi) : r \in \mathcal{R}_w^K, s \in \mathcal{S}_r \cup \{\emptyset\}, \chi \in \Xi_{s,k}, e_\ell - E_\ell^{rem} \geq e^{res}, \forall \ell\}. \quad (6)$$

1.2.5 Charging stations

The station set $\mathcal{S}$ collects service nodes embedded in the road network. Station s is characterised by plug count $\bar{B}_s$, maximum charging power $\bar{p}_s^{ch}$, maximum discharging power $\bar{p}_s^{dis}$, queue capacity $\bar{Q}_s$, dwell capacity $\bar{H}_s$, and grid-bus mapping $\text{bus}(s)$.

1.3 Modelling Assumptions

- A1 (LWR kinematic wave).** Road links follow first-order traffic flow theory (LWR) with triangular FD (2).
- A2 (FIFO).** Vehicles on the same link satisfy first-in-first-out: if $k_{i,a}^{in} \leq k_{j,a}^{in}$ then $k_{i,a}^{out} \leq k_{j,a}^{out}$.
- A3 (Equal PCE).** ICVs and EVs have identical passenger-car equivalence: $\text{PCE}^F = \text{PCE}^E = 1$. Class-specific PCE may be added as an extension.
- A4 (CTM/LTM for propagation).** CTM and LTM handle traffic propagation only; vehicle route choice and station activity are determined by the vehicle behaviour layer.
- A5 (ICV minimum-time).** The base ICV objective is minimum travel time. Fuel costs may be included in the generalised cost as an extension.
- A6 (EV economic optimum).** EVs minimise a generalised cost that includes travel time, station waiting, charging service time, energy price, battery degradation, and SOC feasibility (Section 1.5).
- A7 (Fleet representation).** Vehicles sharing the same OD, departure step, and behaviour type are aggregated into a fleet flow $x_{w,r,k}^F$ or $x_{w,\omega,k}^E$ (Section 1.6).
- A8 (Discrete SOC dynamics).** EV battery energy follows the discrete update (15).
- A9 (Mobility feasibility).** At every step: $e_{v,k} - E_{v,k}^{rem} \geq e_v^{res}$.
- A10 (Sequential charge/discharge).** Charging and discharging are sequential phases of the station visit. An activity plan ω specifies either a charging session ($E_\omega^{ch} \geq 0, E_\omega^{dis} = 0$), a discharging (V2G) session ($E_\omega^{ch} = 0, E_\omega^{dis} \geq 0$), or both as consecutive phases. As shown in Proposition 1 below, any plan with simultaneously positive $p_{v,k}^{ch} > 0$ and $p_{v,k}^{dis} > 0$ at the same step k is strictly dominated in cost under the no-arbitrage price condition (26), so no binary mode variables are required in the typical operating regime.
- A11 (Station as service node).** EVs traverse arrival $\rightarrow$ queue $\rightarrow$ service $\rightarrow$ dwell $\rightarrow$ egress inside the network loading.
- A12 (Plug capacity).** $B_{s,k} \leq \bar{B}_s$ at every step.

1.4 Microscopic Vehicle Model

1.4.1 ICV microscopic state and route choice

ICV v has state $x_{v,k}^F = (l_{v,k}, p_v)$ comprising its current location and chosen path. For OD w departing at step k , the route-choice problem is

$$r_{v,k}^* = \arg \min_{r \in \mathcal{R}_w} T_{r,k}, \quad (7)$$

where the path travel time is computed by dynamic recursion along path $r = (a_1, a_2, \dots, a_m)$. This equation gives individual route choice in *fixed-departure mode* where departure step k is exogenous. In full DUE, ICVs jointly choose departure step and route to minimise $\Psi_{w,r,k}^F$ (which includes schedule delay; see §1.11.2).

$$k_{j+1} = k_j + \left\lceil \frac{T_{a_j, k_j}}{\Delta t} \right\rceil, \quad T_{r,k} = \sum_{j=1}^m T_{a_j, k_j}. \quad (8)$$

Under Wardrop time-equilibrium in *fixed-departure mode* the ICV travel-time condition is

$$T_{r,k} \begin{cases} = T_{w,k}^{\min} & \text{if } x_{w,r,k}^F > 0, \\ \geq T_{w,k}^{\min} & \text{if } x_{w,r,k}^F = 0. \end{cases} \quad (9)$$

Here $x_{w,r,k}^F = h_{w,r,k}^F$ is the equilibrium route flow at fixed departure step k . The generalised-cost condition for *full DUE* (with departure-time endogeneity and schedule delay) is given by (81)–(82) in §1.11.

Fuel consumption and emissions. Link-level fuel consumption for ICV v on link a is

$$F_{v,a,k} = L_a \left(\alpha_F + \frac{\beta_F}{\bar{v}_{a,k}} \right), \quad (10)$$

where α_F (L/km) is the distance-proportional coefficient, β_F (L·h/km) the congestion-sensitive coefficient, and $\bar{v}_{a,k} = L_a/T_{a,k}$ the mean link speed. Tailpipe CO₂ emissions are $G_{v,a,k} = \varepsilon^F F_{v,a,k}$.

1.4.2 EV microscopic state

EV v has a four-dimensional state

$$x_{v,k}^E = (l_{v,k}, e_{v,k}, \sigma_{v,k}, p_v), \quad (11)$$

where $l_{v,k}$ is its location, $e_{v,k}$ its battery energy (kWh), $\sigma_{v,k}$ its behavioural state, and p_v its activity plan. The behavioural state belongs to

$$\sigma_{v,k} \in \{ \text{driving, accessing_station, queueing, charging, discharging, dwelling, egressing, completed} \}. \quad (12)$$

1.4.3 EV driving energy consumption

Link-level energy consumption is

$$E_{v,a}^{\text{drive}} = \xi_v L_a \quad (\text{distance-linear model}), \quad (13)$$

or, with speed dependence,

$$E_{v,a,k}^{\text{drive}} = g_v(L_a, T_{a,k}, \bar{v}_{a,k}). \quad (14)$$

The mean link speed $\bar{v}_{a,k} = L_a/T_{a,k}$ links congestion directly to energy consumption.

1.4.4 EV SOC dynamics

The battery energy evolves as

$$e_{v,k+1} = e_{v,k} - E_{v,k}^{drive} + \eta_v^{ch} p_{v,k}^{ch} \Delta t - \frac{p_{v,k}^{dis} \Delta t}{\eta_v^{dis}}, \quad (15)$$

subject to $e_v^{min} \leq e_{v,k} \leq e_v^{max}$ and the *mobility feasibility* constraint

$$e_{v,k} - E_{v,k}^{rem} \geq e_v^{res}, \quad (16)$$

where $E_{v,k}^{rem}$ is the energy required from the current position to the trip destination and e_v^{res} is the reserve threshold. Equation (16) is the critical coupling between V2G dispatch and mobility: discharging reduces the feasible plan set and may necessitate a second charging stop.

1.4.5 EV charging trigger

An EV *must* seek a charging stop if the safety constraint (16) would be violated without one:

$$e_{v,k} - E_{v,k}^{rem} < e_v^{res}. \quad (17)$$

With economic optimisation the trigger extends to any plan that reduces total cost:

$$\min_{\omega \in \Omega_{v,k}^{stop}} C_{v,\omega,k}^E < C_{v, \text{no stop}, k}^E, \quad (18)$$

meaning an EV with adequate SOC may still enter a station if low energy prices or V2G revenue make a stop economically attractive.

1.5 EV Economic Cost Function

The total generalised cost of activity plan ω for OD w departing at step k is

$$C_{w,\omega,k}^E = C_{w,\omega,k}^{travel} + C_{w,\omega,k}^{wait} + C_{w,\omega,k}^{service} + C_{w,\omega,k}^{energy} + C_{w,\omega,k}^{deg} + C_{w,\omega,k}^{schedule} + \phi_{w,\omega,k}^{soc}. \quad (19)$$

Travel-time cost. $C_{w,\omega,k}^{travel} = \alpha_t T_{\omega,k}$, where α_t (\$/h) is the value of time.

Waiting-time cost. If ω includes station s , with k_s^{arr} the arrival step:

$$C_{w,\omega,k}^{wait} = \alpha_w W_{s,k_s^{arr}}, \quad (20)$$

where $W_{s,k}$ is the expected queue waiting time at station s at step k . This term is zero if $s = \emptyset$.

Service-time cost. Charging and discharging are sequential: a vehicle first charges for duration T_{ω}^{ch} , then (if V2G) discharges for duration T_{ω}^{dis} , occupying a plug for the total time $T_{\omega}^{service} = T_{\omega}^{ch} + T_{\omega}^{dis}$:

$$T_{\omega}^{ch} = \frac{E_{\omega}^{ch}}{\eta_{\omega}^{ch} \bar{p}_s^{ch}}, \quad T_{\omega}^{dis} = \frac{E_{\omega}^{dis}}{\bar{p}_s^{dis}}, \quad (21)$$

and $C_{w,\omega,k}^{service} = \alpha_s T_{\omega}^{service}$.

Remark 2 (Energy-side convention). *Throughout this section, $p_{v,k}^{ch}$ and $p_{v,k}^{dis}$ are grid-side power quantities. The battery receives $\eta_v^{ch} p_{v,k}^{ch} \Delta t$ energy per charging step and releases $p_{v,k}^{dis} \Delta t / \eta_v^{dis}$ per discharging step, consistent with the SOC update (15). The battery-side charged energy and grid-side discharged energy for an activity plan are*

$$E_{\omega}^{ch} = \sum_{\ell \in \mathcal{K}_{\omega}^s} \eta_{\omega}^{ch} p_{\omega,\ell}^{ch} \Delta t, \quad E_{\omega}^{dis} = \sum_{\ell \in \mathcal{K}_{\omega}^s} p_{\omega,\ell}^{dis} \Delta t. \quad (22)$$

Charging payments are based on grid-side energy drawn ($\sum_{\ell} p_{\omega,\ell}^{ch} \Delta t = E_{\omega}^{ch} / \eta^{ch}$); V2G revenue is based on grid-side energy delivered (E_{ω}^{dis}). This convention is consistent with the service-time formula (21) and the energy-cost term (23).

Because discharging extends the total plug-occupancy time $T^{service}$, it competes directly with queueing vehicles and enters the cost function without any binary mode variable.

Energy cost. Let $\pi_{s,\ell}^{ch}$ and $\pi_{s,\ell}^{dis}$ be the per-kWh charging price and discharging compensation at station s during service step ℓ . The net energy cost is

$$C_{w,\omega,k}^{energy} = \sum_{\ell \in \mathcal{K}_{\omega}^s} (\pi_{s,\ell}^{ch} p_{\omega,\ell}^{ch} - \pi_{s,\ell}^{dis} p_{\omega,\ell}^{dis}) \Delta t, \quad (23)$$

where charging is a positive cost and V2G revenue is negative.

Battery degradation cost. A linear equivalent-throughput model gives $C_{w,\omega,k}^{deg} = c^{deg}(E_{\omega}^{ch} + E_{\omega}^{dis})$, where c^{deg} is interpreted as an equivalent cost per kWh of mixed battery/grid-side throughput. If degradation is evaluated strictly on the battery side, replace the discharging term by $E_{\omega}^{dis} / \eta^{dis}$: $C_{w,\omega,k}^{deg} = c^{deg,ch} E_{\omega}^{ch} + c^{deg,dis} E_{\omega}^{dis} / \eta^{dis}$. A cycle-depth model $f^{deg}(\text{DoD}_{\omega}, E_{\omega}^{throughput})$ may replace either form for higher fidelity.

Schedule delay cost. With desired arrival time t_w^* :

$$C_{w,\omega,k}^{schedule} = \gamma_e \max(0, t_w^* - t_{w,\omega,k}^{arr}) + \gamma_l \max(0, t_{w,\omega,k}^{arr} - t_w^*). \quad (24)$$

SOC infeasibility penalty.

$$\phi_{w,\omega,k}^{soc} = M \cdot \mathbf{1}[\exists \ell : e_{\omega,\ell} - E_{\omega,\ell}^{rem} < e^{res}], \quad (25)$$

where M is a large penalty. In optimisation formulations the mobility feasibility (16) is imposed as a hard constraint instead.

Proposition 1 (Sequential charge/discharge under no-arbitrage prices). *Assume the effective round-trip V2G compensation does not exceed the charging cost after round-trip efficiency and degradation:*

$$\pi_{s,k}^{dis} \eta^{ch} \eta^{dis} \leq \pi_{s,k}^{ch} + c^{deg}. \quad (26)$$

Then for any activity plan ω with $p_{\omega,k}^{ch} > 0$ and $p_{\omega,k}^{dis} > 0$ at the same step k , there exists a plan ω' with strictly lower cost: $C_{w,\omega',k}^E < C_{w,\omega,k}^E$.

Proof. Consider any step k at which $p_{\omega,k}^{ch} > 0$ and $p_{\omega,k}^{dis} > 0$. The net battery-energy increment induced by the original plan at this step is

$$\Delta e_k = \eta^{ch} p_{\omega,k}^{ch} \Delta t - \frac{p_{\omega,k}^{dis} \Delta t}{\eta^{dis}}. \quad (27)$$

Construct an alternative plan ω' that achieves the same net SOC increment Δe_k but avoids simultaneous charging and discharging. If $\Delta e_k \geq 0$, set

$$p_{\omega',k}^{ch} = \frac{\Delta e_k}{\eta^{ch} \Delta t}, \quad p_{\omega',k}^{dis} = 0. \quad (28)$$

If $\Delta e_k < 0$, set

$$p_{\omega',k}^{ch} = 0, \quad p_{\omega',k}^{dis} = -\frac{\Delta e_k \eta^{dis}}{\Delta t}. \quad (29)$$

In both cases one verifies directly that $\eta^{ch} p_{\omega',k}^{ch} \Delta t - p_{\omega',k}^{dis} \Delta t / \eta^{dis} = \Delta e_k$, so ω' preserves the same SOC trajectory at step k .

Because the original plan simultaneously charges and discharges, it contains an internal cycling component that strictly increases total energy throughput and plug-occupancy time relative to ω' . Therefore $C_{\omega'}^{service} < C_{\omega}^{service}$ and $C_{\omega'}^{deg} < C_{\omega}^{deg}$.

Under the no-arbitrage condition (26), the eliminated cycling component cannot produce a net cost saving after accounting for round-trip efficiency and degradation, so the energy-cost term does not increase. Therefore $C_{w,\omega',k}^E < C_{w,\omega,k}^E$, proving that no cost-minimising plan contains simultaneous charging and discharging at the same step. $\square$

Remark 3 (Price condition). *Proposition 1 requires the no-arbitrage condition (26): the effective round-trip V2G revenue $\pi_{s,k}^{dis} \eta^{ch} \eta^{dis}$ must not exceed the charging cost $\pi_{s,k}^{ch}$ plus degradation c^{deg} . When this condition fails (strong V2G premium), simultaneous charge/discharge may be cost-improving; binary mode constraints $I_{v,k}^{ch} + I_{v,k}^{dis} \leq 1$ should then be added. Under the no-arbitrage condition, binary variables are not needed and the problem remains a pure LP.*

1.5.1 EV equilibrium condition (fixed-departure special case)

In fixed-departure mode, the EV Wardrop condition at a given step k is:

$$h_{w,\omega,k}^E > 0 \Rightarrow C_{w,\omega,k}^E = C_{w,k}^{E,\min} := \min_{\omega' \in \Omega_{w,k}} C_{w,\omega',k}^E. \quad (30)$$

This is the restriction of the full DUE condition (83)–(84) to a single departure step; the OD-level minimum μ_w^E reduces to the step-level minimum $C_{w,k}^{E,\min}$ when the departure profile is fixed.

1.6 Fleet Aggregation Model

1.6.1 ICV departure-rate variables and demand conservation

The ICV DUE variable is $h_{w,r,k}^F \geq 0$: the number of ICVs of OD w departing at step k and choosing route r . The *full-DUE* demand constraint sums across the departure window:

$$\sum_{k \in \mathcal{K}_w^{dep}} \sum_{r \in \mathcal{R}_w} h_{w,r,k}^F = D_w^F \quad \forall w. \quad (31)$$

In *fixed-departure mode* the per-step constraint replaces (31):

$$\sum_{r \in \mathcal{R}_w} h_{w,r,k}^F = d_{w,k}^F \quad \forall w, k. \quad (32)$$

The notation $x_{w,r,k}^F$ used in earlier sections (route-flow after equilibration) equals $h_{w,r,k}^F$ at the fixed point.

1.6.2 EV activity-plan departure-rate variables and demand conservation

The EV DUE variable is $h_{w,\omega,k}^E \geq 0$: the number of EVs of OD w departing at step k and choosing activity plan ω . The *full-DUE* demand constraint is:

$$\sum_{k \in \mathcal{K}_w^{dep}} \sum_{\omega \in \Omega_{w,k}} h_{w,\omega,k}^E = D_w^E \quad \forall w. \quad (33)$$

In *fixed-departure mode*:

$$\sum_{\omega \in \Omega_{w,k}} h_{w,\omega,k}^E = d_{w,k}^E \quad \forall w, k. \quad (34)$$

1.6.3 Fleet-to-link inflow mapping

Total link inflow $u_{a,k}$ is the sum of contributions from ICV routes, EV activity plans, and station egress flows:

$$u_{a,k} = u_{a,k}^F + u_{a,k}^E + u_{a,k}^{egress}, \quad (35)$$

where dynamic-incidence indicators $\delta_{a,w,r,k}^F$ and $\delta_{a,w,\omega,k}^E$ specify whether a given fleet flow enters link a at step k ; these are determined endogenously by the CTM/LTM propagation layer.

1.6.4 Fleet-to-station arrival and departure

Station s arrival flow:

$$u_{s,k}^S = \sum_w \sum_{\substack{\omega \in \Omega_{w,k} \\ s \in \omega}} \delta_{s,w,\omega,k}^{arr} x_{w,\omega,k}^E, \quad (36)$$

and station departure flow:

$$o_{s,k} = \sum_w \sum_{\substack{\omega \in \Omega_{w,k} \\ s \in \omega}} \delta_{s,w,\omega,k}^{dep} x_{w,\omega,k}^E. \quad (37)$$

The departure flow $o_{s,k}$ re-enters the road network as $u_{a,k}^{egress}$ on the station's egress link.

1.6.5 Class-specific cumulative curves

Class-specific cumulative entry and exit counts sum to the aggregate:

$$N_{a,k}^{in} = \sum_{c \in \mathcal{C}} N_{a,k}^{in,c}, \quad N_{a,k}^{out} = \sum_{c \in \mathcal{C}} N_{a,k}^{out,c}. \quad (38)$$

Link occupancy at step k : $n_{a,k} = N_{a,k}^{in} - N_{a,k}^{out}$.

1.7 CTM-Based System-Level Network Loading

The *Cell Transmission Model* (CTM, Daganzo 1994) discretises each link into a sequence of cells, resolving within-link shockwave dynamics and supporting route-commodity tracking for DUE. The implementation resides in `src/ev_power_traffic/ctm_propagation.cpp`.

1.7.1 Cell discretisation and CFL condition

Link a is partitioned into M_a cells of equal length

$$\delta_a = \frac{L_a}{M_a}, \quad M_a = \max\left(1, \left\lfloor \frac{L_a}{v_a^{ff} \Delta t} \right\rfloor\right). \quad (39)$$

When $L_a \geq v_a^{ff} \Delta t$, the choice in (39) yields $\delta_a = L_a/M_a \geq v_a^{ff} \Delta t$, so the **CFL condition** $\delta_a \geq v_a^{ff} \Delta t$ is satisfied by construction. If $L_a < v_a^{ff} \Delta t$, no cell partition with $M_a \geq 1$ can satisfy this condition; the global time step must be reduced or sub-stepping must be used before applying the CTM. For the full network:

$$\Delta t \leq \min_{a \in \mathcal{A}^R} \frac{\delta_a}{v_a^{ff}}. \quad (40)$$

The jam occupancy per cell is

$$N_{a,m}^{jam} = k_a^j \delta_a = \frac{\bar{q}_a(v_a^{ff} + w_a)}{v_a^{ff} w_a} \delta_a. \quad (41)$$

1.7.2 Cell sending and receiving functions

$$S_{a,m,k} = \min\left(\frac{v_a^{ff}}{\delta_a} n_{a,m,k}, \bar{q}_a\right) \Delta t, \quad (42)$$

$$R_{a,m,k} = \min\left(\bar{q}_a, \frac{w_a}{\delta_a} (N_{a,m}^{jam} - n_{a,m,k})\right) \Delta t. \quad (43)$$

1.7.3 Inter-cell flow and occupancy update

The flow from cell m to cell $m + 1$:

$$y_{a,m \rightarrow m+1,k} = \min(S_{a,m,k}, R_{a,m+1,k}), \quad (44)$$

and the cell occupancy update:

$$n_{a,m,k+1} = n_{a,m,k} + y_{a,m-1 \rightarrow m,k} - y_{a,m \rightarrow m+1,k}. \quad (45)$$

1.7.4 Route-commodity tracking

For multi-route networks a per-route fraction is maintained:

$$\phi_{a,m,k}^r = \frac{n_{a,m,k}^r}{n_{a,m,k} + \varepsilon}, \quad n_{a,m,k} = \sum_r n_{a,m,k}^r, \quad (46)$$

and the per-route inter-cell transfer is $y_{a,m \rightarrow m+1,k}^r = \phi_{a,m,k}^r y_{a,m \rightarrow m+1,k}$.

Remark 4 (Proportional mixing approximation). Equation (46) propagates route identities by proportional mixing: each route receives a share of the inter-cell flow proportional to its current fraction of cell occupancy. This preserves aggregate link conservation but does not enforce route-level FIFO ordering. Vehicles from later-departing routes may be assigned to earlier exit slots. Exact route-specific FIFO tracking requires per-commodity cumulative curves and is handled by the LTM layer (Section 1.8).

1.7.5 Inter-link node model

At node i , the inter-link flow from link a to link b is

$$y_{a \rightarrow b,k} = \beta_{a,b,k} \min(S_{a,k}^{exit}, R_{b,k}^{entry}), \quad (47)$$

where $S_{a,k}^{exit} = S_{a,M_a-1,k}$, $R_{b,k}^{entry} = R_{b,0,k}$, and $\beta_{a,b,k}$ is the turning fraction derived from route fractions or given as exogenous input.

Remark 5 (Node model limitation). Equation (47) is a single-input/single-output approximation valid when each junction has at most one upstream link. For general merge nodes (multiple incoming links sharing one outgoing link), the individual terms $\beta_{a,b,k} \min(S_{a,k}^{exit}, R_{b,k}^{entry})$ may sum to exceed $R_{b,k}^{entry}$, violating the downstream receiving constraint. A proper node model maximises joint throughput subject to sending and receiving constraints simultaneously (see equations (54)–(55) in the LTM section), applying supply-side priority rules at diverge nodes.

Remark 6 (Current CTM implementation scope). The current `ctm_propagation.cpp` implementation applies equation (47) with precomputed turning fractions. Support for general diverge/merge node solvers (simultaneous multi-link capacity allocation) is a planned extension and is not yet enabled.

1.7.6 CTM-based Dynamic User Equilibrium (DUE)

The CTM is embedded in a DUE outer loop. Two solution methods are available:

MSA (heuristic baseline).

$$x^{(i+1)} = x^{(i)} + \frac{1}{i+1} (\hat{x}^{(i)} - x^{(i)}), \quad (48)$$

where $\hat{x}^{(i)}$ is the all-or-nothing assignment to shortest-time routes given CTM travel times from iteration i . The relative gap

$$\epsilon_i = \frac{\sum_r (T^r - T^{\min}) x_r^{(i)}}{\sum_r T^{\min} x_r^{(i)}} \quad (49)$$

drives convergence.

LP-based system-optimal formulation. Section 1.17.1 gives the SO-CTM LP that replaces MSA when system-optimal objectives are acceptable and optimality certificates are required.

Remark 7 (CTM vs LTM granularity). *CTM has $O(M_a|\mathcal{A}|T)$ state variables per simulation; LTM has $O(|\mathcal{A}|T)$. Table 1 in §1.15.7 gives the full trade-off.*

1.8 LTM-Based Meso-Micro Network Loading

The *Link Transmission Model* (LTM, Yperman 2007) propagates traffic using cumulative entry and exit curves rather than per-cell densities. It is the primary propagation layer for EV meso-micro coupling because it provides exact FIFO trajectory reconstruction at $O(|\mathcal{A}|T)$ cost.

1.8.1 Cumulative curves and occupancy

For link a , define cumulative entry and exit counts:

$$u_{a,k} = N_{a,k+1}^{\text{in}} - N_{a,k}^{\text{in}}, \quad v_{a,k} = N_{a,k+1}^{\text{out}} - N_{a,k}^{\text{out}}. \quad (50)$$

Link occupancy: $n_{a,k} = N_{a,k}^{\text{in}} - N_{a,k}^{\text{out}}$.

1.8.2 Free-flow and backward-wave delays

$$\tau_a^{\text{ff}} = \left\lceil \frac{L_a}{v_a^{\text{ff}} \Delta t} \right\rceil, \quad \tau_a^{\text{bw}} = \left\lceil \frac{L_a}{w_a \Delta t} \right\rceil. \quad (51)$$

Remark 8 (Initial history curves). *The sending and receiving functions (52)–(53) reference $N_{a,k-\tau_a^{\text{ff}}}^{\text{in}}$ and $N_{a,k-\tau_a^{\text{bw}}}^{\text{out}}$ for $k < \tau$. For an empty initial network: $N_{a,k}^{\text{in}} = N_{a,k}^{\text{out}} = 0$ for all $k \leq 0$. Non-zero pre-history (e.g., background demand already en route) must be supplied externally before the first simulation step.*

1.8.3 LTM sending and receiving functions

$$S_{a,k} = \min \left(N_{a,k-\tau_a^{\text{ff}}}^{\text{in}} - N_{a,k}^{\text{out}}, \bar{q}_a \Delta t \right), \quad (52)$$

$$R_{a,k} = \min \left(N_{a,k-\tau_a^{\text{bw}}}^{\text{out}} + N_a^{\text{jam}} - N_{a,k}^{\text{in}}, \bar{q}_a \Delta t \right). \quad (53)$$

The boundary flow constraints are $0 \leq v_{a,k} \leq S_{a,k}$ and $0 \leq u_{a,k} \leq R_{a,k}$.

Remark 9 (Discrete-offset convention). *In the current implementation, entry flows at step k are written to $N_{a,k+1}^{in}$ (flows are increments between steps), and exit flows are written after the sending calculation. Depending on the update ordering, the first observable exit index in the code may appear at $k + \tau_a^{ff} + 1$ or $k + \tau_a^{ff} + 2$, rather than the theoretical $k + \tau_a^{ff}$. This is an implementation-indexing convention arising from the entry-increment and outflow write-back ordering, not an intrinsic LTM property. The theoretical minimum travel time remains $\tau_a^{ff} \Delta t$.*

1.8.4 LTM node model

For intersection node i with incoming link set $\mathcal{A}^{in}(i)$ and outgoing link set $\mathcal{A}^{out}(i)$:

$$\sum_{b \in \mathcal{A}^{out}(i)} y_{a,b,k} \leq S_{a,k}, \quad a \in \mathcal{A}^{in}(i), \quad (54)$$

$$\sum_{a \in \mathcal{A}^{in}(i)} y_{a,b,k} \leq R_{b,k}, \quad b \in \mathcal{A}^{out}(i). \quad (55)$$

Link flows satisfy

$$v_{a,k} = \sum_{b \in \mathcal{A}^{out}(i)} y_{a,b,k}, \quad (56)$$

$$u_{b,k} = \sum_{a \in \mathcal{A}^{in}(i)} y_{a,b,k} + d_{b,k}^{orig} + o_{s \rightarrow b,k}, \quad (57)$$

where $d_{b,k}^{orig}$ is exogenous origin demand and $o_{s \rightarrow b,k}$ is the egress flow entering link b from station s .

1.8.5 FIFO-consistent vehicle trajectory reconstruction

The LTM layer is mesoscopic and does not explicitly store individual trajectories. They are recovered by FIFO-consistent cumulative-count inversion.

For a vehicle entering link a during step k , its FIFO rank lies in the cumulative interval

$$\xi_{v,a} \in (N_{a,k}^{in}, N_{a,k+1}^{in}]. \quad (58)$$

If vehicles are represented individually, the j -th vehicle entering during step k is assigned $\xi_{v,a} = N_{a,k}^{in} + j$; for continuous flows a representative rank within the interval is used. The exit step is then

$$k_{v,a}^{out} = \min\{k : N_{a,k}^{out} \geq \xi_{v,a}\}. \quad (59)$$

Repeating over the activity path yields the full trajectory: $\mathcal{T}_v = \{(a, k_{v,a}^{in}, k_{v,a}^{out}) : a \in p_v\}$.

Station ranks are assigned analogously in cumulative intervals, where $U_{s,k}$ is cumulative arrivals, $B_{s,k}^{cum}$ cumulative service starts, $C_{s,k}^{cum}$ cumulative completions, and $O_{s,k}$ cumulative departures at station s by step k : arrivals during step k occupy $(U_{s,k}, U_{s,k+1}]$, service starts occupy $(B_{s,k}^{cum}, B_{s,k+1}^{cum}]$, and completions occupy $(C_{s,k}^{cum}, C_{s,k+1}^{cum}]$. For the vehicle arriving as the $\xi_{v,s}^{arr}$ -th at station s :

$$k_{v,s}^{start} = \min\{k : B_{s,k}^{cum} \geq \xi_{v,s}^{arr}\}, \quad (60)$$

$$k_{v,s}^{end} = \min\{k : C_{s,k}^{cum} \geq \xi_{v,s}^{start}\}, \quad (61)$$

$$k_{v,s}^{dep} = \min\{k : O_{s,k} \geq \xi_{v,s}^{end}\}. \quad (62)$$

This links aggregate LTM states to individual EV records without a full microscopic simulation.

1.9 Charging Stations as Endogenous Service Nodes

1.9.1 Station state and dynamics

Station s tracks three aggregate populations: $Q_{s,k}$ (queueing), $B_{s,k}$ (in service), $H_{s,k}$ (post-service dwell). The state transitions are

$$Q_{s,k+1} = Q_{s,k} + u_{s,k}^S - b_{s,k}, \quad (63)$$

$$B_{s,k+1} = B_{s,k} + b_{s,k} - c_{s,k}, \quad (64)$$

$$H_{s,k+1} = H_{s,k} + c_{s,k} - o_{s,k}. \quad (65)$$

1.9.2 Service admission

$$b_{s,k} \leq Q_{s,k} + u_{s,k}^S, \quad B_{s,k} + b_{s,k} \leq \bar{B}_s. \quad (66)$$

1.9.3 Service completion

$$c_{s,k} \leq B_{s,k}, \quad c_{s,k} \leq \mu_{s,k}^{step}, \quad (67)$$

where in the fluid station model

$$\mu_{s,k}^{step} = \frac{\bar{B}_s \Delta t}{\bar{T}_{s,k}^{service}} \quad (68)$$

is the per-step service completion capacity (veh/step). In the individual-vehicle mode, $c_{s,k} = \sum_v \mathbf{1}[k_{v,s}^{end} = k]$.

1.9.4 Station departure

$$o_{s,k} \leq H_{s,k} + c_{s,k}, \quad o_{s,k} \leq R_{s,k}^{egress}, \quad (69)$$

where $R_{s,k}^{egress}$ is the receiving capacity of the downstream road link. An overcrowded downstream link prevents EV departure even after service completion.

1.9.5 Queue spillback to road

If $\bar{Q}_s < \infty$ is the queue capacity, the access-link receiving capacity is reduced as the queue fills:

$$R_{s,k}^{access} = R_{s,k}^0 \left(1 - \frac{Q_{s,k}}{\bar{Q}_s} \right)_+, \quad (70)$$

so that $R_{s,k}^{access} = 0$ when $Q_{s,k} \geq \bar{Q}_s$. This feedback propagates station congestion back into the road network.

1.10 Charging and V2G Feedback on Traffic

Charging and V2G decisions affect traffic through three channels.

Channel 1: Route and plan choice. Low energy prices or high V2G compensation may induce EVs to choose a longer route that passes a favoured station:

$$C_{\omega_B,k}^E < C_{\omega_A,k}^E \quad \Rightarrow \quad x_{w,\omega_B,k}^E > 0, \quad (71)$$

changing link inflows $u_{a,k}$ and thus congestion.

Channel 2: SOC depletion and secondary charging. V2G discharge reduces $e_{v,k}$, potentially triggering the safety condition (17) later in the trip. A secondary charging stop adds an extra station visit, increases $u_{s',k}^S$ at a downstream station s' , and extends total trip time.

Channel 3: Dwell time and egress. Charging duration $T_\omega^{ch} = E_\omega^{ch}/(\eta^{ch}\bar{p}_s^{ch})$ directly determines how long a vehicle occupies a plug. Larger charging energy increases $B_{s,k}$, reduces $\bar{B}_s - B_{s,k}$, and can grow the queue $Q_{s,k}$, which eventually activates the spillback mechanism (70).

Summary of coupling.

$$\underbrace{(E^{ch}, E^{dis}, \pi^{ch}, \pi^{dis})}_{\text{charging/V2G strategy}} \longrightarrow \underbrace{(u_{s,k}^S, o_{s,k}, u_{a,k})}_{\text{station and link flows}} \longrightarrow \underbrace{(N^{in}, N^{out}, T_{a,k})}_{\text{LTM/CTM state}} \longrightarrow \underbrace{C_{w,\omega,k}^E}_{\text{EV cost feedback}}. \quad (72)$$

1.11 Dynamic Equilibrium Model

This module supports two assignment modes: (i) **Full DUE**: departure time, route, and EV activity plan are endogenous choices under OD total-demand constraints (31)–(33); equilibrium costs are compared across all departure steps and alternatives. (ii) **Fixed-departure DUE**: the departure profile $d_{w,k}$ is given; only route/activity-plan choice is equilibrated within each departure step under constraints (32)–(34). Unless stated otherwise, DUE refers to mode (i).

1.11.1 Dynamic Network Loading operator

The DNL operator maps departure-rate flows to network states:

$$\mathcal{D} : (h^F, h^E) \mapsto (T, W, Q, o, N^{in}, N^{out}). \quad (73)$$

1.11.2 Experienced-cost operators

After DNL produces network states, the *experienced-cost operators* convert states to generalised path costs:

$$\Psi_{w,r,k}^F(h) := \alpha_t T_{w,r,k}^{exp}(h) + C_{w,r,k}^{schedule,F}(h), \quad (74)$$

$$\Psi_{w,\omega,k}^E(h) := C_{w,\omega,k}^E(h), \quad (75)$$

where $T_{w,r,k}^{exp}(h)$ is the FIFO-reconstructed experienced travel time of a departure- k vehicle on route r , $C_{w,\omega,k}^E$ is the EV generalised cost from (??), and the ICV schedule-delay penalty is

$$C_{w,r,k}^{schedule,F}(h) = \gamma_e \max(0, t_w^* - t_{w,r,k}^{arr}) + \gamma_l \max(0, t_{w,r,k}^{arr} - t_w^*), \quad (76)$$

with $t_{w,r,k}^{arr} = k\Delta t + T_{w,r,k}^{exp}$, desired arrival time t_w^* , and early/late penalties $\gamma_e, \gamma_l \geq 0$. Setting $\gamma_e = \gamma_l = 0$ recovers the travel-time-only special case.

1.11.3 ICV and EV choice operators

The DUE minimum costs are OD-level minima across *all* departure steps and alternatives in the departure window:

$$\mu_w^F := \min_{k \in \mathcal{K}_w^{dep}, r \in \mathcal{R}_w} \Psi_{w,r,k}^F(h), \quad (77)$$

$$\mu_w^E := \min_{k \in \mathcal{K}_w^{dep}, \omega \in \Omega_{w,k}} \Psi_{w,\omega,k}^E(h). \quad (78)$$

The ICV assignment operator $\mathcal{A}^F$ and EV assignment operator $\mathcal{A}^E$ map experienced costs to departure-rate flows. Under all-or-nothing (AON) assignment for full DUE, the entire OD demand is assigned to the minimiser across all (k, r) (resp. (k, ω)):

$$\hat{h}_{w,r,k}^F = \begin{cases} D_w^F & (r, k) \in \arg \min_{r', k'} \Psi_{w,r',k'}^F, \\ 0 & \text{otherwise,} \end{cases} \quad (79)$$

$$\hat{h}_{w,\omega,k}^E = \begin{cases} D_w^E & (\omega, k) \in \arg \min_{\omega', k'} \Psi_{w,\omega',k'}^E, \\ 0 & \text{otherwise.} \end{cases} \quad (80)$$

Ties are broken by uniform splitting. A logit stochastic model gives $P_{w,\omega,k}^E \propto \exp(-\theta \Psi_{w,\omega,k}^E)$ and is normalised so that $\sum_{k,\omega} h_{w,\omega,k}^E = D_w^E$.

In *fixed-departure mode*, the minimisation in (77)–(80) is restricted to the given departure step k , and D_w^F, D_w^E are replaced by $d_{w,k}^F, d_{w,k}^E$.

1.11.4 Wardrop DUE conditions

At a full DUE h^* , no vehicle can reduce its experienced cost by unilaterally changing departure time, route, or activity plan:

$$h_{w,r,k}^{F,*} > 0 \Rightarrow \Psi_{w,r,k}^F(h^*) = \mu_w^{F,*}, \quad (81)$$

$$h_{w,r,k}^{F,*} = 0 \Rightarrow \Psi_{w,r,k}^F(h^*) \geq \mu_w^{F,*}, \quad (82)$$

$$h_{w,\omega,k}^{E,*} > 0 \Rightarrow \Psi_{w,\omega,k}^E(h^*) = \mu_w^{E,*}, \quad (83)$$

$$h_{w,\omega,k}^{E,*} = 0 \Rightarrow \Psi_{w,\omega,k}^E(h^*) \geq \mu_w^{E,*}. \quad (84)$$

Here $\mu_w^{F,*}$ and $\mu_w^{E,*}$ are the equilibrium (minimised) OD costs from (77)–(78) evaluated at h^* .

Remark 10 (Relationship to fixed-departure condition). *The condition in (30) is the fixed-departure special case of (83)–(84) with the minimum taken only over ω at a given k .*

1.11.5 Fixed-point equilibrium

The coupled system is at equilibrium when

$$(h^{F,*}, h^{E,*}) = \mathcal{A}(\mathcal{D}(h^{F,*}, h^{E,*})), \quad (85)$$

i.e. given the congestion and queues produced by the current departure rates, no vehicle has an incentive to change its departure time, route, or activity plan.

1.12 Solution Algorithm: MSA for Mixed Fleet DUE

Starting from initial flows $h^{F,(0)}, h^{E,(0)}$ satisfying the demand constraints, each iteration i proceeds as:

1. **DNL forward pass:** $(T^{(i)}, W^{(i)}, Q^{(i)}, o^{(i)}) = \mathcal{D}(h^{F,(i)}, h^{E,(i)})$.
2. **Compute experienced costs:** Evaluate $\Psi_{w,r,k}^{F,(i)}$ and $\Psi_{w,\omega,k}^{E,(i)}$ via (74)–(75) for all (w, r, k) and (w, ω, k) .
3. **AON assignment (cross- k):** For full DUE, assign each OD's total demand to the single minimiser across all departure steps and alternatives:

$$\hat{h}_{w,r,k}^{F,(i)} = \begin{cases} D_w^F & (r, k) \in \arg \min_{r', k'} \Psi_{w,r',k'}^{F,(i)}, \\ 0 & \text{otherwise,} \end{cases} \quad (86)$$

and analogously for $\hat{h}_{w,\omega,k}^{E,(i)}$ with D_w^E . In fixed-departure mode, the minimisation is restricted to each k with demand $d_{w,k}^F$ (resp. $d_{w,k}^E$).

4. **MSA update** (step size $\lambda_i = 1/(i+1)$):

$$h^{F,(i+1)} = h^{F,(i)} + \lambda_i (\hat{h}^{F,(i)} - h^{F,(i)}), \quad h^{E,(i+1)} = h^{E,(i)} + \lambda_i (\hat{h}^{E,(i)} - h^{E,(i)}). \quad (87)$$

5. **Convergence check.** The *ICV relative gap* (OD-level minimum):

$$\epsilon_i^F = \frac{\sum_{w,k,r} h_{w,r,k}^{F,(i)} (\Psi_{w,r,k}^{F,(i)} - \mu_w^{F,(i)})}{\sum_w D_w^F \mu_w^{F,(i)}}, \quad (88)$$

where $\mu_w^{F,(i)} = \min_{k,r} \Psi_{w,r,k}^{F,(i)}$. The *EV relative gap*:

$$\epsilon_i^E = \frac{\sum_{w,k,\omega} h_{w,\omega,k}^{E,(i)} (\Psi_{w,\omega,k}^{E,(i)} - \mu_w^{E,(i)})}{\sum_w D_w^E |\mu_w^{E,(i)}| + \varepsilon}, \quad (89)$$

where $\mu_w^{E,(i)} = \min_{k,\omega} \Psi_{w,\omega,k}^{E,(i)}$ and $\varepsilon > 0$ guards against a near-zero denominator when EV costs include V2G revenue that can make $\mu_w^{E,(i)}$ negative. Stop when $\max(\epsilon_i^F, \epsilon_i^E) < \epsilon^{tol}$.

1.13 Key Output Metrics

Traffic metrics. Total travel time in network: $TTT = \sum_{a,k} n_{a,k} \Delta t$. Total system travel time: $TSTT = \sum_{w,r,k} x_{w,r,k}^F T_{w,r,k} + \sum_{w,\omega,k} x_{w,\omega,k}^E T_{w,\omega,k}$. Network throughput: $\sum_{a \in \mathcal{A}_{sink}} N_{a,T}^{out}$.

Station metrics. Maximum queue $Q_s^{max} = \max_k Q_{s,k}$, average waiting time $\bar{W}_s = \sum_k Q_{s,k} \Delta t / \sum_k u_{s,k}^S$, and plug utilisation $\rho_s = \sum_k B_{s,k} / (\bar{B}_s T)$. The step-level expected waiting time used in the EV cost (20) is estimated by the fluid approximation $W_{s,k} \approx Q_{s,k} / \mu_{s,k}^{hour}$, where $\mu_{s,k}^{hour} = \bar{B}_s / \bar{T}_{s,k}^{service}$ (measured in veh/hour when $\bar{T}_{s,k}^{service}$ is in hours) gives $W_{s,k}$ in hours. The equivalent per-step capacity is $\mu_{s,k}^{step} = \mu_{s,k}^{hour} \Delta t = \bar{B}_s \Delta t / \bar{T}_{s,k}^{service}$ (veh/step).

EV energy metrics. Total charging energy $E_{tot}^{ch} = \sum_{s,k} P_{s,k}^{ch} \Delta t$, total V2G energy $E_{tot}^{dis} = \sum_{s,k} P_{s,k}^{dis} \Delta t$, net load energy $E_{tot}^{net} = E_{tot}^{ch} - E_{tot}^{dis}$, and V2G potential:

$$P_{s,k}^{v2g,pot} = \sum_{v \in \mathcal{V}_{s,k}^{dwell}} \min \left(\bar{p}_v^{dis}, \frac{e_{v,k} - e_v^{res}}{\Delta t} \eta_v^{dis} \right). \quad (90)$$

Equation (90) gives the *one-step* V2G power potential for vehicles currently dwelling at the station. Multi-period V2G dispatch additionally requires intertemporal constraints on SOC (15) and remaining dwell time $k_{v,s}^{dep} - k$ to ensure feasibility across the full dwell window.

Station net load. Station charging and discharging powers are $P_{s,k}^{ch} = \sum_v p_{v,k}^{ch}$ and $P_{s,k}^{dis} = \sum_v p_{v,k}^{dis}$, giving net load $P_{s,k}^{net} = P_{s,k}^{ch} - P_{s,k}^{dis}$ and bus-level EV injection $P_{b,k}^{EV} = \sum_{s: \text{bus}(s)=b} P_{s,k}^{net}$.

1.14 Correctness Checklist

The following invariants must hold throughout every simulation:

1. **Vehicle conservation.** $N_{a,k}^{out} \leq N_{a,k}^{in}$ for all a, k .
2. **Road capacity.** Admitted link inflows and outflows satisfy $u_{a,k} \leq \bar{q}_a \Delta t$ and $v_{a,k} \leq \bar{q}_a \Delta t$. Exogenous origin demand may exceed this value; any excess is stored in an origin queue and admitted in subsequent steps.

3. **Jam storage.** $n_{a,k} \leq N_a^{jam}$.
4. **Plug capacity.** $B_{s,k} \leq \bar{B}_s$.
5. **SOC feasibility.** $e_v^{min} \leq e_{v,k} \leq e_v^{max}$ and $e_{v,k} - E_{v,k}^{rem} \geq e_v^{res}$.
6. **FIFO.** If $k_{i,a}^{in} \leq k_{j,a}^{in}$ then $k_{i,a}^{out} \leq k_{j,a}^{out}$.
7. **Sequential charge/discharge** (Proposition 1). $p_{v,k}^{ch} \cdot p_{v,k}^{dis} = 0$ at every step k when the no-arbitrage condition (26) holds; sequential behaviour then follows from cost optimality alone, without binary mode variables. When V2G compensation substantially exceeds charging cost after round-trip efficiency, binary mode constraints $I_{v,k}^{ch} + I_{v,k}^{dis} \leq 1$ should be added.

1.15 Numerical Results

The following results are produced by five test suites, all assertions passing:

- `test_ltm_stress.cpp` — 20 cases, 1532 assertions (LTM wave propagation, CFL, conservation).
- `test_ctm_ltm_comparison.cpp` — 5 cases, 118 assertions (CTM vs. LTM throughput and travel-time accuracy).
- `test_ev_power_traffic_report.cpp` — 4 cases, 11 assertions (fleet assignment and V2G simulation with per-step output).
- `test_ev_power_traffic_joint_opt_d.cpp` — 10 cases, 93 assertions (joint social-welfare optimizer, FormD).
- `test_ev_power_traffic_largescale.cpp` — 3 cases, 14 assertions (10,000-EV scalability benchmark on IEEE-118 power grid).

Results are organised in two groups. The first group (§§1.15.2–??) contains **implementation diagnostics**: tests that characterise known numerical artefacts (integer-delay, CFL) and compare CTM vs. LTM behaviour. These are diagnostic results for implementation quality, not validation of the DUE equilibrium model. The second group (§§1.15.8–1.15.15) contains **model validation tests**: fleet assignment, V2G optimality, and scalability results that exercise the DUE and joint-optimisation layers.

1.15.1 Implementation diagnostics

These tests characterise the discrete-time LTM and CTM numerical behaviour; they are not model-validation results for the DUE equilibrium. Production DUE runs should use time steps and link discretisations for which the artefacts quantified below (sustained-capacity degradation, travel-time overestimation at low CFL numbers) are controlled or verified by time-step refinement tests.

1.15.2 Single-link LTM: propagation delay and CFL

For a single urban arterial link ($L = 2$ km, $v^{ff} = 50$ km/h, $w = 16.7$ km/h, $\bar{q} = 900$ veh/h, $\Delta t = 2$ min):

$$v^{ff} \Delta t = 50 \times \frac{1}{30} = 1.67 \text{ km} < L = 2 \text{ km} \quad [\text{CFL satisfied}],$$

$$\tau^{ff} = \lceil \frac{2}{1.67} \rceil = 2, \quad \tau^{bw} = \lceil \frac{2}{0.556} \rceil = 4, \quad N^{jam} \approx 144 \text{ veh}, \quad \text{cap_step} = 30 \text{ veh}.$$

A 20-vehicle pulse (well below $\text{cap_step} = 30$) enters at step 0. The LTM (Group XXI) shows:

- $N^{in}[T] = 20$ (full pulse absorbed),
- $N^{out}[T] = 20$ (all vehicles drain; $T = 30 \gg \tau^{ff} + \tau^{bw} = 6$),
- cumulative curves are monotone non-decreasing for all k ,
- CTM gives identical conservation ($N^{in} = N^{out} = 20$).

1.15.3 Freeway sustained-capacity: LTM vs CTM throughput

Freeway link ($L = 10$ km, $v^{ff} = 100$ km/h, $w = 25$ km/h, $\bar{q} = 2000$ veh/h, $\Delta t = 5$ min):

$$v^{ff} \Delta t = 8.33 \text{ km} < L = 10 \text{ km} \quad [\text{CFL satisfied}],$$

$$\tau^{ff} = 2, \quad \tau^{bw} = 5, \quad N^{jam} = 1000 \text{ veh}, \quad \text{cap_step} = 166.7 \text{ veh}.$$

With $T = 80$ steps of sustained capacity demand (Group XXII):

Model	Exit vehicles	Fraction of $T \cdot \text{cap_step}$
CTM	13 300	99.7%
LTM	8 833	66.2%

The CTM nearly matches $T \cdot \text{cap_step} = 13\,333$; in the current single-cell implementation, the inject-propagate update ordering allows a vehicle entering at step k to exit at the same step (a zero-delay implementation artefact that occurs when too few cells are used). The LTM throughput is lower due to a discretisation-induced storage–delay mismatch: with $\tau^{ff} = 2$ and $\tau^{bw} = 5$, the discrete delay product gives $\bar{q} \Delta t (\tau^{ff} + \tau^{bw}) = 166.7 \times 7 = 1167$ veh, which exceeds $N^{jam} = 1000$ veh. This causes an oscillatory fill/drain pattern. A crude fill/block-cycle approximation gives throughput on the order of $6/(6+5) \approx 55\%$ of capacity; the finite-horizon implementation reports 66.2% because start-up transients, terminal drainage, and the discrete update convention allow additional vehicles to exit before the horizon closes. The key point is not the exact fraction but the presence of a storage–delay mismatch under the coarse $\Delta t = 5$ min discretisation. **This is a numerical artefact** that should be controlled by reducing Δt , using the corrected discrete jam density $N_a^{jam, disc} = \bar{q}_a \Delta t (\tau_a^{ff} + \tau_a^{bw})$, or a corrected update convention — it is not an intrinsic LTM property.

1.15.4 CFL-violation pathology

Intentionally violating CFL by using $L = 0.5$ km with the same freeway parameters ($v^{ff} \Delta t = 8.33 \text{ km} \gg L$, Group XXIII):

Model	Average throughput	Fraction of cap
LTM	150 veh/h	7.5% (collapses)
CTM coarse (same Δt)	2000 veh/h	100% (physically invalid)
CTM fine ($\Delta t_{ctm} = 0.2$ min, CFL OK)	1999 veh/h	100% (correct)

The LTM collapses because $\tau^{ff} = \tau^{bw} = 1$, making $N^{jam} = 50$ while $\text{cap_step} = 166.7$, a fundamental contradiction. The coarse CTM spikes cell occupancies above N^{jam} but drains them in the same step, giving full capacity at a physically meaningless state. The fine CTM with CFL-satisfying sub-steps recovers correct behaviour. **Remedy:** always enforce $L \geq v^{ff} \Delta t$ at the modelling stage.

1.15.5 Four-link arterial corridor

Four copies of the urban arterial link (Group XXIV, $T = 120$):

Model	Exit vehicles	Dead-band	Prediction
LTM	1992	12 steps ($= 4(\tau^{ff} + 1)$)	$(T - 12) \times 30 = 3240$
CTM	3576	3 steps ($= N_L - 1$)	$(T - 3) \times 30 = 3510$

The LTM dead-band is the sum of per-link discrete offsets; the CTM dead-band is one step per inter-link crossing (zero-delay per link). LTM exits 1992 vehicles $\approx 61\%$ of its ceiling (same discretisation-induced oscillatory throughput phenomenon as Group XXII; each link contributes a $\tau^{bw} = 4$ blocking cycle).

1.15.6 Travel-time accuracy: LTM offset vs CTM zero-delay

Group XXV tests a 12-vehicle pulse on both a short link ($L = 1$ km, $\Delta t = 1$ min) and a freeway link ($L = 10$ km, $\Delta t = 5$ min):

Link	True ff (min)	LTM 1st exit	CTM step-0 exit	LTM overestimate
Short	1.2	step 4 = 4 min	10.0 veh	+233%
Freeway	6.0	step 4 = 20 min	10.0 veh	+233%

Both models have $\tau^{ff} = 2$; the LTM discrete-offset convention adds two more steps ($\tau^{ff} + 2 = 4$), placing the true delay between the CTM zero-lower-bound and the LTM upper-bound.

1.15.7 CTM vs LTM: model comparison

Table 1: CTM vs LTM: structural and computational comparison.

Feature	CTM (Daganzo 1994)	LTM (Yperman 2007)
State variable	Cell occupancy $n_{a,m,k}$	Cumulative curves $N_{a,k}^{in}, N_{a,k}^{out}$
State size	$O(M_a \mathcal{A} T)$	$O(\mathcal{A} T)$
CFL	$\delta_a \geq v_a^{ff} \Delta t$ (explicit)	$\tau^{ff} = \lceil L / (v^{ff} \Delta t) \rceil$ (implicit)
Travel delay	Zero-delay artefact in single-cell implementation	$\tau^{ff} \Delta t$ in theory; current implementation adds 1–2 indexing steps
FIFO consistency	Approximate (cell-averaged)	Exact (cumulative-count inversion)
Backward wave	Cell-to-cell propagation	Exact τ^{bw} look-back on N^{out}
Oscillatory throughput	Not observed in tested CFL-satisfying multi-cell cases	Possible under coarse time steps when discrete storage and delay are mismatched
Route tracking	Per-cell commodity fractions	Per-class cumulative curves
LP formulation size	$O((\sum_a M_a + \mathcal{R}) T)$	$O(\mathcal{A} T)$
Individual EV coupling	Via commodity fraction reconstruction	Direct FIFO inversion of N^{out}
Typical use	System-level DUE, shockwave analysis	Meso-micro EV-charging coupling
Implementation	<code>ctm_propagation.cpp</code>	<code>ltm_propagation.cpp</code>

1.15.8 Model validation tests

1.15.9 EV penetration sweep

A single station ($\bar{B} = 10$ plugs, mean service $\bar{\mu}^{-1} = 0.50$ h) on a 1-link corridor; 40% of EVs require charging each step: Queue lengths grow linearly with penetration. At 70% the arrival

Table 2: EV penetration sweep: station loading.

EV penetration	Max queue (veh)	Avg plug utilisation
10%	449	60.3%
30%	1 936	60.6%
50%	3 426	60.6%
70%	4 915	60.6%

rate ($1500 \times 0.7 \times 0.4 = 420$ veh/h) exceeds service capacity ($10/0.5 = 20$ veh/h) by factor 21, confirming the runaway queue prediction.

1.15.10 Individual EV trajectories and V2G

Three EV agents traverse a 3-link corridor (6.5 km total, $T = 96$ steps, $\Delta t = 0.25$ h): No EV

Table 3: Individual EV agent profiles. Battery = 80 kWh, consumption = 0.20 kWh/km, reserve = 8 kWh.

Agent	Initial SOC	Initial energy (kWh)	Final SOC
EV-high	85%	68.0	83.4%
EV-mid	45%	36.0	43.4%
EV-low	20%	16.0	18.4%

triggered the safety condition (17). For the most vulnerable agent (EV-low, $e_0 = 16$ kWh), the remaining energy after traversing the full 6.5 km corridor is $e_0 - E^{drive} = 16.0 - 6.5 \times 0.20 = 14.7$ kWh; since $14.7 > e^{res} = 8$ kWh, no charging stop is triggered. FIFO trajectory reconstruction (59) verified for a 20-vehicle cohort: all 20 trajectories satisfy temporal consistency, $k^{out} \geq k^{in}$, minimum travel time $\geq \tau^{ff} \Delta t$, and strict FIFO order.

1.15.11 Fleet assignment: station-capacity benchmark (4-node, 5 EVs)

Network and parameters. A 4-node, 4-link network with two routes from node 1 to node 4 ($\Delta t = 1$ h, $T = 6$, default station price 0.30 \$/kWh). Five EVs each request 20 kWh of charging energy:

- **Route 1** (near): links 11→12; free-flow 0.50 h; station 101 at node 2 — 40 kW total, 2 fast plugs at 20 kW/veh.
- **Route 2** (far): links 21→22; free-flow 1.00 h; station 102 at node 3 — 60 kW total, 3 fast plugs at 20 kW/veh.

All 5 vehicles have the same origin/destination (node 1→4) departing at step 0. Table 4 compares three assignment algorithms.

Remark 11 (SO/greedy vs. DUE-MSA). *DSP, Greedy, and SO-LP are single-iteration assignment methods, not DUE equilibrium solvers. They demonstrate energy delivery correctness and*

Table 4: Fleet assignment comparison: 4-node benchmark, 5 EVs. h_r : route flow; E^{del} : delivered energy; E^{unsv} : unserved energy; VOT = 1.0 \$/h. Source: `test_ev_power_traffic_report.cpp` (11 assertions, all passing).

Algorithm	h_1 (veh)	h_2 (veh)	E^{del} (kWh)	E^{unsv} (kWh)	Travel (veh·h)	LP obj (\$)
DSP	5	0	40.0	60.0	2.50	—
Greedy	2	3	100.0	0.0	4.00	—
SO-LP	2	3	100.0	0.0	4.00	34.00

station-capacity enforcement, not Wardrop optimality. The CTM-MSA DUE solver is validated separately in Section 1.15.12 (4-node, 4 EVs, `test_ev_power_traffic_ctm_due.cpp`, 12 test cases, 520 assertions passing), which reports final EV Wardrop gap ϵ^E , ICV gap ϵ^F , session synthesis, energy delivery, schedule-delay route reversal, and price-of-anarchy.

DSP station overload. The Deterministic Shortest Path assigns all 5 vehicles to the near route, ignoring station capacity. Station 101 can supply only 40 kW (2 plugs $\times$ 20 kW/plug), so 2 of 5 vehicles receive their full 20 kWh each (= 40 kWh delivered) and 3 vehicles receive nothing (= 60 kWh unserved). A queue forms: 5 vehicles are waiting in queue at step 2 (average wait 1.75 h), 3 at step 3 (1.05 h), 1 at step 4 (0.35 h), confirming the $M/D/c$ -type queue dynamics of Section 1.9.

Greedy and SO-LP agreement. Both the Capacity-Aware Greedy and the HiGHS-certified System-Optimal LP produce the identical split ($h_1 = 2$, $h_2 = 3$), saturating both stations at exactly their installed capacity (40 and 60 kW respectively) for one step each with zero queue. Table 5 shows the per-step station states for the Greedy case; the SO-LP results are numerically identical.

Table 5: Per-step station state — Capacity-Aware Greedy and SO-LP (identical). P_s^{ch} : station charging power (kW); B_s : vehicles actively charging; H_s : vehicles in post-service dwell awaiting egress. At step 2, $B = 0$ and $H > 0$ because all charging completed at step 1; no pre-service waiting ($Q_s = 0$ throughout).

Step k	P_{101}^{ch} (kW)	B_{101} (veh)	H_{101} (veh)	P_{102}^{ch} (kW)	B_{102} (veh)	H_{102} (veh)
0	0	0	0	0	0	0
1	40	2	0	60	3	0
2	0	0	2	0	0	3
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0

System-Optimal LP objective. The LP minimises total system cost; with station price 0.30 \$/kWh and VOT = 1.0 \$/h:

$$J^{SO} = \underbrace{0.30 \times 100}_{\text{energy cost}=\$30.00} + \underbrace{1.0 \times (2 \times 0.5 + 3 \times 1.0)}_{\text{travel delay}=\$4.00} = \$34.00. \quad (91)$$

The LP is proven optimal by HiGHS (NativeDualSimplex backend), MIP gap = 0 (pure LP, no integer variables), primal infeasibility $< 10^{-14}$.

1.15.12 CTM-DUE MSA validation: 4-node benchmark

Setup. The same 4-node, 4-link symmetric network of Section 1.15.11 is used, but solved with the full CTM-MSA DUE algorithm (eq:ctm-due-algorithm). 4 EVs depart at step 0, each requesting 10 kWh at station 101 (route 1) or station 102 (route 2); $\Delta t = 1$ h, $T = 5$, $\epsilon^{tol} = 10^{-4}$, max 50 iterations. Source: `test_ev_power_traffic_ctm_due.cpp` (12 test cases, 520 assertions, all passing; B1–B3 outputs below).

B1: Session synthesis and dispatch.

Metric	Value
Converged	yes
MSA iterations	1
ϵ^E (final EV Wardrop gap, eq:ev-gap)	0.00×10^0
ϵ^F (final ICV Wardrop gap, eq:icv-gap)	0.00×10^0
Sessions synthesised	1
E^{req} (total requested energy)	40.00 kWh
E^{del} (total delivered energy)	40.00 kWh
E^{unshr} (unserved energy)	0.00 kWh
Service ratio	100.0%
Route 1 flow h_1	4.000 veh
Route 2 flow h_2	0.000 veh
Peak station 101 load (step 1)	42.1 kW
Station 101 queue	0 veh (all steps)

On an uncongested symmetric network with no schedule delay all routes have equal generalised cost at the initial equal-split, so the relative gap $\epsilon^E = 0$ at iteration 1. The auxiliary AoN assignment concentrates flow on route 1 (tie-broken by index); the MSA step produces $h_1 = 4$, $h_2 = 0$. All 4 vehicles are served without queueing ($W_{s,k} = 0$ hr for all s, k), and the single synthesised session delivers all 40 kWh.

B2: SOC feasibility filter. Route 1 requires $2 \times 0.5 \text{ km} \times 2 \text{ kWh/km} = 2.0 \text{ kWh}$ driving energy; route 2 requires $2 \times 1.5 \times 2 = 6.0 \text{ kWh}$. With $e_0 = 2.0 \text{ kWh}$, route 2 is infeasible (eq:soc-feasibility-filter) and is removed before AoN assignment: $h_1 = 4.0000$ veh, $h_2 = 0.000000$ veh. Total flow conserved: $h_1 + h_2 = 4.0$.

B3: Price-of-anarchy benchmark. $\Delta t = 0.1$ h, $T = 6$; both DUE and SO benchmarks run.

Metric	Value
Converged	yes
MSA iterations	1
ϵ^E (final EV gap)	0.00×10^0
DUE TSTT	0.4000 veh·hr
SO TSTT	0.0000 veh·hr
PoA ratio	1.00
SO benchmark solved	yes

At free-flow speed on an uncongested network the DUE TSTT equals $4 \text{ veh} \times 0.10 \text{ hr/route} = 0.40 \text{ veh} \cdot \text{hr}$. The SO benchmark produces TSTT = 0 because the system-optimal LP minimises total link flow weighted by travel time; with zero congestion (all BPR terms at base value) the LP finds a trivial lower bound of 0. The PoA ratio = DUE TSTT/SO TSTT is reported as 1.0 when the denominator is below 10^{-9} (both values are near free-flow; no congestion to exploit).

Schedule-delay reversal (test case 11). To validate eq:psi-icv and eq:icv-schedule-delay, the asymmetric network (route 1: ff ≈ 0.10 hr; route 2: ff ≈ 0.30 hr) is run with an ICV demand (6 vehicles) whose desired arrival time is $t^* = 0.25$ hr and early penalty $\gamma_e = 20$ \$/hr, $\gamma_l = 0$. Without schedule delay, route 1 wins (shorter travel time). With the penalty:

$$\Psi_1^F = 0.10 + \frac{\gamma_e \max(0, 0.25 - 0.10)}{VOT} = 0.10 + 20 \times 0.15 = 3.10 \$, \quad \Psi_2^F = 0.30 + 0 = 0.30 \$.$$

AoN assigns all 6 ICV vehicles to route 2 (verified: $h_2 > h_1$, total = 6.0 veh, $\epsilon^F \geq 0$ and finite).

Gap fields (test case 12). On a symmetric EV-only problem converged to $\epsilon^{tol} = 10^{-3}$: $\epsilon^F = 0$ (no ICV demands), $\epsilon^E \geq 0$, finite, and $\epsilon^E < \epsilon^{tol}$ upon convergence; `CTMDUEResult::relative_gap` = $\max(\epsilon^E, \epsilon^F)$ (all verified by 4 assertions).

1.15.13 Joint social welfare optimizer: FormD small case (2-bus, 4 EVs)

Network and parameters. A 2-bus DC power system (bus 1: generator G1 at 10 \$/MWh, $\bar{p}_1^g = 2$ MW; bus 2: generator G2 at 50 \$/MWh, $\bar{p}_2^g = 0.5$ MW; branch 1–2: $x = 0.1$ pu, rate = 0.5 MW). Traffic sub-network identical to Section 1.15.11 (4 nodes, 2 routes). 4 EVs, WTP = \$50/veh, energy request = 10 kWh/veh, VOT = 1.0 \$/h, station price = 0.03 \$/kWh; $\Delta t = 1$ h, $T = 6$. Station 101 on bus 1 (40 kW, 2 plugs); station 102 on bus 2 (60 kW, 3 plugs).

Table 6 reports three solver modes from `test_ev_power_traffic_joint_opt_d.cpp` (93 assertions, all passing).

Table 6: FormD joint optimizer: 2-bus, 4-EV results. SWmax = SocialWelfareMax; UBM = UserBenefitMax; Vars/Cons: LP variable/constraint counts; W : social welfare.

Mode	h_1 (veh)	h_2 (veh)	B_{EV} (\$)	C_{del} (\$)	C_{gen} (\$)	W (\$)	Vars / Cons
SWmax, no OPF	4.0	0.0	200.00	2.00	0.00	198.00	11 / 27
SWmax + DC-OPF	4.0	0.0	200.00	2.00	0.40	197.60	53 / 45
UBM, no OPF	4.0	0.0	200.00	2.00	0.00	—	11 / 27

Welfare accounting. All four vehicles prefer the near route: WTP = \$50/veh exceeds both the travel delay ($0.5 \text{ h} \times \$1/\text{h} = \$0.50/\text{veh}$) and the energy cost ($10 \text{ kWh} \times 0.03 \$/\text{kWh} = \$0.30/\text{veh}$). Station 101 delivers $4 \times 10 = 40$ kWh in one step at $P_{101}^{ch} = 4 \times 10 \text{ kW} = 40 \text{ kW}$.

$$B_{EV} = 4 \times \$50 = \$200. \quad (92)$$

$$C_{del} = \text{VOT} \times (4 \times 0.5 \text{ h}) = \$2.00. \quad (93)$$

$$C_{gen} = 0 \quad (\text{no OPF coupling in SWmax no-OPF mode}). \quad (94)$$

$$W = B_{EV} - C_{del} - C_{gen} = 200 - 2 - 0 = \$198.00. \quad (95)$$

With DC-OPF coupling, generator G1 dispatches 0.04 MW at step 1 to balance the EV load:

$$C_{gen} = 10 \text{ \$/MWh} \times 0.04 \text{ MW} \times 1 \text{ h} = \$0.40, \quad W = 200 - 2 - 0.40 = \$197.60. \quad (96)$$

The DC-OPF formulation adds 53 LP variables (generators, voltage angles, branch flows for all T steps) and 45 constraints versus the 11 variables and 27 constraints of the traffic-only LP.

For UserBenefitMax the LP minimises $C_{del} - B_{EV} + C_{elec}$ where $C_{elec} = 4 \times 10 \text{ kWh} \times 0.03 \text{ \$/kWh} = \$1.20$, giving objective = $2.00 - 200.00 + 1.20 = -\196.80 .

Solver certificate. All three modes: proven optimal, solver status `Optimal`, NativeDualSimplex backend, primal infeasibility $\leq 7 \times 10^{-11}$, integrality violation = 0 (pure LP).

1.15.14 V2G pure-LP validation (Proposition 1)

Setup. One EV ($n_{vehicles} = 1$) on route 1 only; $e_0 = 50 \text{ kWh}$, $e_{min} = 20 \text{ kWh}$, $e_{max} = 60 \text{ kWh}$; dwell = 2 steps; station price $\pi_s = 1.0 \text{ \$/kWh}$ (high, to incentivise maximal discharge); $\bar{p}^{dis} = 10 \text{ kW}$, $\bar{p}^{ch} = 20 \text{ kW}$; WTP = \\$50; VOT = $1.0 \text{ \$/h}$; mode: UserBenefitMax with V2G enabled. No binary variables are present (pure LP following Proposition 1).

Remark 12 (No-arbitrage condition in this test). *This test does not rely on Proposition 1 to enforce sequential behaviour. The EV begins with sufficient SOC and the objective rewards only delivered V2G energy; charging is never economically attractive, so the LP solution is discharge-only. The no-arbitrage condition (26) is not the reason no simultaneous charging occurs in this particular instance—it is a consequence of the LP cost structure under these specific parameters. The no-arbitrage condition $\pi^{dis} \eta^{ch} \eta^{dis} \leq \pi^{ch} + c^{deg}$ is a price-parameter condition: when it holds, a LP solution with $p_{v,k}^{ch} p_{v,k}^{dis} = 0$ is provably at least as good as any simultaneous schedule (Proposition 1). If instead $\pi^{dis} \eta^{ch} \eta^{dis} > \pi^{ch} + c^{deg}$ (arbitrage is profitable), binary mode constraints $I_{v,k}^{ch} + I_{v,k}^{dis} \leq 1$ would be required.*

Optimal V2G plan. Table 7 shows the LP-optimal per-step charging schedule.

Table 7: V2G pure-LP: per-step session plan for 1 EV (`test_ev_power_traffic_joint_opt_d.cpp`, test D9, all assertions passing).

Dwell step k	$p_{v,k}^{ch}$ (kW)	$p_{v,k}^{dis}$ (kW)	$e_{v,k\text{-start}}$ (kWh)	$e_{v,k\text{-end}}$ (kWh)	$p^{ch} \cdot p^{dis}$	Prop. 1
1	0.00	10.00	50	40	0	✓
2	0.00	10.00	40	30	0	✓

The product $p_{v,k}^{ch} \cdot p_{v,k}^{dis} = 0$ at both steps: the EV discharges at maximum 10 kW for both dwell steps without any simultaneous charging. Under the no-arbitrage condition (26) (satisfied here: $\pi^{dis} \eta^{ch} \eta^{dis} = 1.0 \times 1 \times 1 \leq \pi^{ch} + c^{deg} = \infty$), the LP optimum is provably non-simultaneous by Proposition 1, so no binary mode constraints are required *for this test instance*. If the no-arbitrage condition were violated, binary constraints $I_{v,k}^{ch} + I_{v,k}^{dis} \leq 1$ would be needed.

Objective decomposition. Total V2G energy = $2 \times 10 \text{ kW} \times 1 \text{ h} = 20 \text{ kWh}$; V2G revenue = $20 \text{ kWh} \times \$1.0/\text{kWh} = \20 ; travel delay $C_{del} = 1.0 \times 0.5 \text{ h} = \0.50 :

$$J^{UBM} = C_{del} - B_{EV} - E_{tot}^{dis} \cdot \pi_s = 0.50 - 50.00 - 20.00 \cdot 1.0 = -\$69.50. \quad (97)$$

Verified by test assertion: `res.objective = -69.50` (tolerance 10^{-6}); `proven_optimal=true`; primal infeasibility $< 10^{-7}$.

Simulation cross-check. Test Scenario 4 in `test_ev_power_traffic_report.cpp` independently simulates one V2G session ($e_0 = 50$ kWh, $p^{dis} = 20$ kW, 1 dwell step): final SOC = 30 kWh, $E_{tot}^{dis} = 20$ kWh, EV power balance margin increases by 0.020 MW at the V2G step. Both the simulation test and the joint-optimizer LP agree on $E^{dis} = 20$ kWh.

1.15.15 Large-scale benchmark: 10×10 grid, 10,000 EVs, IEEE-118

System description.

- **Power grid:** IEEE 118-bus; buses 1–100 each host a fast-charging station with 300 fast plugs at 50 kW each (15 MW/station, \$0.30/kWh).
- **Traffic grid:** 10×10 Cartesian grid, 100 nodes, 180 directed links; free-flow 0.25 h/link, capacity 2000 veh/h.
- **Corridors:** 5 north–south corridors, each with two routes. Route A: 9 vertical links down the left column, 1 horizontal link at the bottom (charging stop at mid-left node, row 4). Route B: 1 horizontal link at the top, 9 vertical links down the right column (charging stop at mid-right node, row 4). $\Rightarrow$ 10 routes total.
- **Demands:** 5 corridors $\times$ 4 departure steps = 20 fleet demands, 500 vehicles/demand = 10 000 vehicles total. Each vehicle requests 30 kWh at up to 50 kW.
- $\Delta t = 1$ h, $T = 12$; enforce road capacity and station capacity.

Algorithm comparison. Table 8 reports results from `test_ev_power_traffic_largescale.cpp` (14 assertions, all passing).

Remark 13 (Scalability vs. DUE validation). *The algorithms compared here (CapacityAwareGreedy and SystemOptimalLP) are assignment heuristics and a system-optimal LP, respectively; they are not DUE-MSA runs. This benchmark demonstrates scalability (10,000 EVs, 100 stations, IEEE-118 power grid) and energy-delivery correctness, not equilibrium convergence. A DUE-MSA run on this network would additionally report: final ICV gap ϵ^F , final EV gap ϵ^E , number of MSA iterations, departure-time split, and used-path cost dispersion. These metrics are omitted here because the test uses a single-iteration greedy/SO-LP approach for performance benchmarking.*

Table 8: Large-scale benchmark: 10×10 grid, 10,000 EVs, IEEE-118. All 10,000 vehicles served, $E^{del} = 300,000$ kWh, $E^{unsrv} = 0$.

Algorithm	Solve time (ms)	Travel time (veh·h)	Peak station (kW)	LP obj	Proved optimal
CapacityAwareGreedy	0.5	25,001	9,000	—	no
SystemOptimalLP	3.1	25,015	15,000	115,000	yes

Route assignment. Under CapacityAwareGreedy each of the 20 demands splits 300/200 (Route A/Route B), distributing load between mid-left stations (41, 43, 45, 47, 49) and mid-right stations (42, 44, 46, 48, 50). Under SystemOptimalLP all 500 vehicles per demand concentrate on Route A (LP objective = 115,000), loading left-column stations at their 15 MW installed capacity but achieving certified global optimality.

Table 9: Per-station utilisation under CapacityAwareGreedy. Route-A stations (odd IDs) serve 300 vehicles/demand; Route-B stations (even IDs) serve 200 vehicles/demand; each station active for 4 departure-step groups.

Station (bus)	Route	Peak load (kW)	Delivered energy (kWh)
41	A, corridor 0	9,000	36,000
42	B, corridor 0	6,000	24,000
43	A, corridor 1	9,000	36,000
44	B, corridor 1	6,000	24,000
45	A, corridor 2	9,000	36,000
46	B, corridor 2	6,000	24,000
47	A, corridor 3	9,000	36,000
48	B, corridor 3	6,000	24,000
49	A, corridor 4	9,000	36,000
50	B, corridor 4	6,000	24,000
Total	—	—	300,000

Station utilisation (CapacityAwareGreedy). Table 9 reports peak load and total delivered energy for the 10 route stations across all 4 departure steps.

Energy balance verified: $10,000 \times 30 \text{ kWh} = 300,000 \text{ kWh}$, matching `total_delivered_energy_kwh` to within 1 kWh. Route-A peak: each vehicle requests 30 kWh in a 1-hour step, requiring an average charging power of $30 \text{ kWh}/1 \text{ h} = 30 \text{ kW}$ (within the 50 kW plug hardware limit); $300 \text{ veh} \times 30 \text{ kW/veh} = 9,000 \text{ kW}$. Route-B analogously: $200 \times 30 = 6,000 \text{ kW}$. Under SystemOptimalLP all 500 vehicles in each demand window are concentrated on Route A ($500 \times 30 = 15,000 \text{ kW}$, exactly saturating the 15 MW station capacity).

Scalability. Both solvers complete well within their declared time limits (CapacityAwareGreedy: 0.5 ms vs. 5 s limit; SystemOptimalLP: 3.1 ms vs. 30 s limit), demonstrating that the LP and greedy heuristic both scale efficiently to 10,000-vehicle fleets on a 100-node grid.

1.16 Extension Interfaces to Power and Resilience Models

Output quantity	Consuming module
Station charging power $p_{s,k}^{ch}$	Smart-charging / DC-OPF (Section ??)
Station dwell population $H_{s,k}$ and per-EV SOC	V2G aggregator $\rightarrow$ OPF
EV departure flow $o_{s,k}$	LTM downstream receiving update
Aggregate load $P_{b,k}^{EV}$	Power-network load profile
V2G potential $P_{s,k}^{v2g,pot}$	Reserve and resilience models

1.17 Optimization Framework

The simulation sub-modules use MSA for DUE and rule-based heuristics for charging admission. This section gives proper mathematical programmes that replace or wrap those heuristic layers, providing optimality guarantees and enabling joint power-traffic problems.

1.17.1 CTM-based system-optimal traffic assignment (SO-CTM LP)

Under minimisation of total vehicle-time the non-smooth $\min(\cdot, \cdot)$ in (44) is linearised by two one-sided upper bounds (Ziliaskopoulos 2000).

Objective.

$$\min \sum_{a,m,k} n_{a,m,k} \Delta t. \quad (98)$$

Constraints.

$$n_{a,m,k+1} = n_{a,m,k} + y_{a,m-1,k} - y_{a,m,k} \quad \forall a, m, k \quad (99)$$

$$y_{a,m,k} \leq \frac{v_a^{ff}}{\delta_a} n_{a,m,k} \Delta t \quad \forall a, m, k \quad (100)$$

$$y_{a,m,k} \leq \bar{q}_a \Delta t \quad \forall a, m, k \quad (101)$$

$$y_{a,m,k} \leq \frac{w_a}{\delta_a} (N_{a,m+1}^{jam} - n_{a,m+1,k}) \Delta t \quad \forall a, m < M_a - 1, k \quad (102)$$

$$\sum_{r \in \mathcal{R}(od)} x_{r,k} = d_k^{od} \quad \forall od, k \quad (103)$$

$$n_{a,m,0} = 0 \quad \forall a, m \quad (104)$$

The initial empty condition $n_{a,m,0} = 0$ is standard. The terminal clearance constraint $n_{a,m,T} = 0$ is valid only when the planning horizon T is long enough that all demand has been served; for shorter horizons, replace it with a terminal penalty $\lambda_T \sum_{a,m} n_{a,m,T}$ in the objective (98).

The receiving constraint (102) applies to internal cell-to-cell flows ($m < M_a - 1$). For the exit flow from the last cell $m = M_a - 1$, the downstream receiving capacity is imposed by the node model of the successor link or sink: if link a discharges to a sink, the receiving capacity is set to the sink capacity (or ∞); for an ordinary downstream link the first-cell receiving constraint of that link provides the bound via the CTM node equations. LP size: $O((\sum_a M_a + |\mathcal{R}|)T)$ variables, $O(\sum_a M_a T)$ constraints.

1.17.2 LTM-based system-optimal assignment (SO-LTM LP)

Objective.

$$\min \sum_{a,k} (N_{a,k}^{in} - N_{a,k}^{out}) \Delta t = \min \sum_{a,k} n_{a,k} \Delta t, \quad (105)$$

the total vehicle-hours in the network (minimising total delay). The terminal residual $\sum_a (N_{a,T}^{in} - N_{a,T}^{out})$ may be used as an alternative when the planning horizon T is long enough that the network empties; it is easier to evaluate but ignores travel time throughout the simulation period.

Constraints (all linear).

$$N_{a,k}^{out} - N_{a,k-1}^{out} \leq N_{a,k-\tau_a^{ff}}^{in} - N_{a,k-1}^{out} \quad (\text{sending}) \quad (106)$$

$$N_{a,k}^{out} - N_{a,k-1}^{out} \leq \bar{q}_a \Delta t \quad (\text{cap}) \quad (107)$$

$$N_{a,k}^{in} - N_{a,k-1}^{in} \leq N_{a,k-\tau_a^{bw}}^{out} + N_a^{jam} - N_{a,k-1}^{in} \quad (\text{receiving}) \quad (108)$$

$$N_{a,k}^{in} - N_{a,k-1}^{in} \leq \bar{q}_a \Delta t \quad (\text{cap}) \quad (109)$$

$$\sum_b y_{a,b,k} = N_{a,k}^{out} - N_{a,k-1}^{out} \quad (\text{exit split}) \quad (110)$$

$$\sum_a y_{a,b,k} + d_{b,k}^{orig} = N_{b,k}^{in} - N_{b,k-1}^{in} \quad (\text{entry merge}) \quad (111)$$

$$N_{a,k}^{in} \geq N_{a,k-1}^{in}, \quad N_{a,k}^{out} \geq N_{a,k-1}^{out} \quad (\text{monotone}) \quad (112)$$

LP size: $O(|\mathcal{A}|T)$, independent of cell count.

1.17.3 DUE as a variational inequality

Let the feasible departure-rate set be

$$\mathcal{H} = \left\{ (h^F, h^E) \geq 0 : \sum_{k \in \mathcal{K}_w^{dep}} \sum_r h_{w,r,k}^F = D_w^F, \sum_{k \in \mathcal{K}_w^{dep}} \sum_\omega h_{w,\omega,k}^E = D_w^E, \forall w \right\}. \quad (113)$$

The full-DUE feasible set $\mathcal{H}$ from (113) contrasts with the fixed-departure feasible set:

$$\mathcal{H}^{fix} = \left\{ (h^F, h^E) \geq 0 : \sum_r h_{w,r,k}^F = d_{w,k}^F, \sum_\omega h_{w,\omega,k}^E = d_{w,k}^E, \forall w, k \right\}. \quad (114)$$

The mixed-fleet DUE condition (81)–(84) is equivalent to the variational inequality: find $(h^{F,*}, h^{E,*}) \in \mathcal{H}$ such that

$$\sum_{w,r,k} \Psi_{w,r,k}^F(h^*) (h_{w,r,k}^F - h_{w,r,k}^{F,*}) + \sum_{w,\omega,k} \Psi_{w,\omega,k}^E(h^*) (h_{w,\omega,k}^E - h_{w,\omega,k}^{E,*}) \geq 0 \quad \forall (h^F, h^E) \in \mathcal{H}. \quad (115)$$

For static or separable link-cost approximations the VI can be reduced to the Beckmann potential

$$\min_{(h^F, h^E) \in \mathcal{H}} \Phi(h) = \sum_a \int_0^{f_a(h)} t_a(u) du \quad (116)$$

which is convex when link cost functions are monotone and separable, and can be solved by Frank-Wolfe or interior-point methods. In the full dynamic CTM/LTM setting with spillback and station queues, path costs Ψ^F, Ψ^E depend on departure time and link costs are non-separable; the DUE is therefore treated as the fixed-point problem (85) or the VI (115) rather than a Beckmann program. The Beckmann formulation and its uniqueness certificate are available only under the separable-monotone static approximation.

1.17.4 Joint power-traffic social welfare optimisation

$$\max_{x, p^g, p^{ch}} \sum_d W_d^P s_d - \sum_g C_g(p^g) - \text{VOT} \sum_r x_r T^r(x), \quad (117)$$

subject to DC power balance, LTM traffic dynamics, and charging-load coupling. The general coupling is expressed through station net load:

$$p_{b,k}^{EV} = \frac{1}{1000} \sum_{s: \text{bus}(s)=b} (P_{s,k}^{ch} - P_{s,k}^{dis}) \quad [\text{MW}], \quad (118)$$

where

$$P_{s,k}^{ch} = \sum_{\omega \ni s} x_{\omega,k}^E p_{\omega,k}^{ch}, \quad P_{s,k}^{dis} = \sum_{\omega \ni s} x_{\omega,k}^E p_{\omega,k}^{dis}. \quad (119)$$

Remark 14 (P^{EV} sign convention). *In this traffic module, positive $p_{b,k}^{EV}$ denotes net EV load (aggregate charging exceeds discharging). In power-flow equations using injection-positive convention, EV load enters as $-p_{b,k}^{EV}$, so that generation equals load plus net EV demand.*

In simplified benchmark cases without V2G and with all admitted vehicles charging at a fixed plug power, this reduces to

$$p_{b,k}^{EV} = \frac{1}{1000} \sum_{s: \text{bus}(s)=b} \sum_{r \ni s} x_{r,k} \bar{p}_s^{ch}. \quad (120)$$

The current benchmark tests use this simplified form; the station net-load form (118) is the mathematical interface for full V2G coupling. The `hacdcpf` library solves this via iterative

price-based decomposition (`joint_social_welfare.cpp`): (i) traffic step with SO-LTM LP and current electricity prices; (ii) DC-OPF step with current EV load; (iii) LMP price update with dampening; (iv) convergence check on max price change. This structure alternates between the traffic and power subproblems using primal EV-load signals and dual marginal-cost price signals; a full Dantzig-Wolfe column-generation formulation with a master problem and pricing subproblems is a planned extension.

1.17.5 MILP for charging infrastructure design

Let $z_s \in \{0, 1\}$ and $\bar{B}_s \in \mathbb{Z}_+$:

$$\min_{z, \bar{B}, x, n} \sum_s (c_s^{cap} \bar{B}_s z_s + c_s^{op} \sum_k p_{s,k}^{ch} \Delta t) + \text{VOT} \sum_r x_r T^r, \quad (121)$$

with $B_{s,k} \leq \bar{B}_s z_s$ and $\bar{B}_s \leq \bar{B}_s^{max} z_s$ (big- M linking constraints), plus CTM/LTM dynamics and OD conservation. The MILP is passed to SCIP via the `engine::MIPModel` interface.

1.17.6 LTM-based model predictive control (MPC)

At each step k , solve a finite-horizon LP over horizon H with stage cost

$$\ell_{k+j} = \omega_Q \sum_s Q_{s,k+j} + \omega_T \sum_a (N_{a,k+j}^{in} - N_{a,k+j}^{out}) - \omega_{V2G} \sum_s P_{s,k+j}^{v2g,pot}, \quad (122)$$

controlled via admission rates $b_{s,k}, \dots, b_{s,k+H-1}$. Only the first-step action $b_{s,k}$ is applied; the LP is resolved at step $k+1$ with the observed state.

Recommended integration sequence.

1. **Offline design:** SO-CTM LP or infrastructure MILP to select station locations, capacities, and base route flows.
2. **Day-ahead:** joint social welfare optimisation (`solve_joint_social_welfare`) to set station prices $\pi_{s,k}$ aligning EV charging with grid marginal costs.
3. **Real-time:** LTM-MPC adjusting admission rates in response to observed queue lengths and grid emergency signals.

Table 10: C++ extension map for optimisation-based traffic models.

Formulation	Source file	Reuses
SO-CTM LP	<code>ctm_so_lp.cpp</code>	<code>engine::LPModel</code> , HiGHS
SO-LTM LP	<code>ltm_so_lp.cpp</code>	<code>assignment_lp.cpp</code>
DUE VI / Beckmann	<code>ctm_due_vi.cpp</code>	Frank-Wolfe with CTM forward pass
Joint SW	<code>joint_social_welfare.cpp</code> (implemented)	SO-LTM LP + DC-OPF
Infrastructure MILP	<code>infra_design_milp.cpp</code>	<code>engine::MIPModel</code> , SCIP
LTM-MPC	<code>ltm_mpc.cpp</code>	<code>ltm_propagation.cpp</code> , SO-LTM LP